

DC499 Report Processor Suite

Developer Handoff Guide

Last updated: July 7, 2026

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GitHub: github.com/dkarim02/DC499-reports | dkarim02.github.io/DC499-reports

1. Current File Versions

File	Version	Notes
Menu_v1.6.html	v1.6	Points to all current dept versions
Ecom_v2_5.html	v2.5	Full Report default, location filter, Push to Dashboard, disclaimer footer
Reserve_v1_7.html	v1.7	Merged Picking tile, putaway units, shift selector, location filter, disclaimer
ItemPrep_v2.0.html	v2.0	Full Report ported from Ecom — ILR + ConditionCode
Receiving_v2.0.html	v2.0	Full Report ported from Ecom — LPN Receive by Container ID
Changelog.html	—	Reads from changelog.json
changelog.json	—	Source of truth for patch notes
MegaDash_v1.2.html	v1.2	Built, not yet integrated — parked until live dashboard solved

All files must be in the same folder. index.html redirects to Menu_v1.6.html.

Filename convention: Ecom/Reserve use underscores (Ecom_v2_5.html). ItemPrep/Receiving use dots (ItemPrep_v2.0.html). Menu openApp() calls must match exactly.

2. Architecture

No build system — pure HTML/CSS/JS, open in any browser. No npm, no bundler.

External CDN (cloudflare): PapaParse 5.4.1 (CSV parsing), Chart.js 4.4.1 (charts)

localStorage Keys

App	Roster	Goals	Theme	Shift
Ecom	sc_roster_2nd_v5 / sc_roster_1st_v1	sc_goals_v1	dc499_ecom_theme_v1	ecom_shift_v1
Reserve	rs_roster_v1	rs_goals_v1	dc499_reserve_theme_v1	res_shift_v1
ItemPrep	ip_roster_2nd_v1 / ip_roster_1st_v1	ip_goals_v1	dc499_itemprep_theme_v1	—
Receiving	recv_roster_v1	recv_goals_v1	dc499_receiving_theme_v1	—

■ Bump storage key version (e.g. v5 to v6) when changing default roster. localStorage is tied to the GitHub Pages domain — rosters persist across file updates.

3. Transaction IDs & Metrics

Ecom v2.5

Section	Transaction IDs	Metric
Replenishment	iLPN Replen Fill, Retail iLPN Replen Pull, iLPN Replen Pull, iLPN Replen Fill Large, iLPN Replen Pull Large	Sum Completed Quantity
Putaway	System Directed Putaway, User Directed Putaway	Count rows
Picking	Ecom Mezz Pick To Putwall Cart, Ecom Non-Mezz Pick To Putwall Cart	Sum Quantity
Packing	NRDR CORE PACK FOR ECOM PACK STATION	Sum Quantity
Shipping	OB Putaway By Ship Via	Sum Quantity
Sorting	OB Sort To Putwall Cubby + Criteria = NRDR_SORT_TO_PUTWALL_CUBBYISSUE + Criteria	Sum Quantity

Reserve v1.7

Section	Transaction IDs	Metric
Pick F1	Non Haz Retail Pick To oLPN Cart	Sum Quantity
Pick F2	Non Haz Retail Pick To oLPN Cart Floor 2	Sum Quantity
Replenishment	iLPN Replen Fill, iLPN Replen Fill Large	Sum Completed Quantity
Putaway	System Directed Putaway, User Directed Putaway	Sum Completed Quantity

Reserve Full Report: Wide Picking tile (F1+F2 combined) spans top. Replen and Putaway split the row below.

ItemPrep v2.0

Section	Transaction IDs	Metric
Item Level Receive	Item Level Receive	Sum Quantity, dedup by Container ID
Condition Code	IlpnConditionCodeRemoval	Sum Quantity, dedup by Container ID

Receiving v2.0

Section	Transaction IDs	Metric
LPN Receive	LPN Level Receive, Small Parcel LPN Level Receive	Count unique Container IDs

4. Location-Based Dept Filter — CRITICAL

Applies to shared transaction IDs only: System/User Directed Putaway, iLPN Replen Fill/Pull variants.

Zone Logic (3rd character of location string)

- Zone H = Reserve Stock
- Any other zone = Ecom
- Priority: Current Location wins. If blank, use Previous Location. If both blank, default to Ecom.

Location Naming Convention

Prefix	Meaning	Example
P	Pallet active location	P1H0528A01 = pallet, floor 1, zone H (Reserve)
R	Reserve (contained iLPN) location	R1B0423 = reserve, floor 1, zone B (Ecom)
F	Shelf active location	F2C0612 = shelf, floor 2, zone C (Ecom)

Ecom zones: A, B, C, D, E, F and others. Reserve zones: H only (all floors).

Applied in 3 places per file: processData(), addToFullReport(), renderSidePanel()

```
function rowBelongsToEcom(r) {
  var txId = (r['Transaction ID'] || '').trim();
  if (SHARED_TX.indexOf(txId) === -1) return true; // not shared
  var curZone = getLocationZone(r['Current Location']);
  if (curZone !== null) return curZone !== 'H';
  var prevZone = getLocationZone(r['Previous Location']);
  if (prevZone !== null) return prevZone !== 'H';
  return true; // both blank – default Ecom
}
```

5. Ecom Full Report Key Variables

```
var SORT_CRITERIA = 'NRDR_SORT_TO_PUTWALL_CUBBIES_CRITERIA';
var FULL_TX = {
  replen: ['iLPN Replen Fill', 'Retail iLPN Replen Pull', 'iLPN Replen Pull', 'iLPN Replen Fill Large', 'iLPN Replen F
  putaway: ['System Directed Putaway', 'User Directed Putaway'],
  picking: ['Ecom Mezz Pick To Putwall Cart', 'Ecom Non-Mezz Pick To Putwall Cart'],
  packing: ['NRDR CORE PACK FOR ECOM PACK STATION'],
  shipping: ['OB Putaway By Ship Via'],
  sorting: ['OB Sort To Putwall Cubby']
};
var FULL_METRIC = {
  replen: 'Completed Quantity', putaway: null, // null = count rows
  picking: 'Quantity', packing: 'Quantity', shipping: 'Quantity', sorting: 'Quantity'
};
var FULL_COLORS = {
  replen: '#185FA5', putaway: '#0F6E56', picking: '#9B4DCA',
  packing: '#E67E22', shipping: '#0E9E8E', sorting: '#C0392B'
};
```

Reserve Full Report Key Variables

```
var RES_TX = {
  pick: ['Non Haz Retail Pick To oLPN Cart'],
  pick2: ['Non Haz Retail Pick To oLPN Cart Floor 2'],
  replen: ['iLPN Replen Fill', 'iLPN Replen Fill Large'],
  putaway: ['System Directed Putaway', 'User Directed Putaway']
};
var RES_METRIC = {pick: 'Quantity', pick2: 'Quantity',
```

```
replen:'Completed Quantity', putaway:'Completed Quantity'};
// Note: putaway switched from null (count) to 'Completed Quantity' (units) in v1.7
```

6. Dedup Strategy — CRITICAL

addToFullReport(rows, fileName) uses cross-file-only dedup:

- Key = Employee + '|' + Transaction ID + '|' + Activity Datetime
- Rows in multiple files: first occurrence kept
- Rows duplicated within same file: ALL kept (legitimate)
- Same filename dropped twice: blocked by fullReportFiles array

■ NEVER use CP Trace Id for dedup — it is shared across multiple legitimate transactions and will silently drop valid rows (caused 723 missing replen units previously).

7. Full Report Render Order

- 1. Header + row count meta
- 2. Shift Summary — clickable dept tiles
- 3. Activity by Hour — multi-line chart, smooth/spiky toggle
- 4. Layout wrapper — dept grid (left) + side panel (right, hidden until tile clicked)

Ecom: 3x2 grid. Reserve: Wide Picking tile top, Replen + Putaway row below.

8. Associate Performance Charts

- 5-minute buckets, only active buckets plotted
 - X-axis labels every 30 minutes only
 - Red segments where gap > 10 minutes
 - Tooltip on red points: gap start, end, duration
 - Bottom right: total gap time summed
 - Charts stored in assocCharts{}, destroyed on collapse
-

9. Shift Selector

Shift	Hours	Bounds
1st shift	3:00 AM - 1:45 PM	3-14
2nd shift	2:15 PM - 1:00 AM	14-25

Saves to localStorage. Updates chart time window, roster label, subtitle, and copy output.

Functions: getShift() / setShift(shift) used throughout both Ecom and Reserve.

10. Fin Fcst PPH Values

Department	Fin Fcst PPH
Replenishment	111.6
Putaway	19.6
Picking	93.1
Packing	24.3
Shipping	245.9
Sorting	81.4

Goal Projection Formula (not yet built)

Active associates x 7.5 hrs x Fin Fcst PPH = Projected shift goal

Rules:

- Shift start: 2:15 PM flat
- Transacting hours: 7.5 (8 hrs minus 30 min lunch)
- Projection cap: 10:45 PM
- If no goal set within first 2 hours: use actively transacting associates (not roster count)
- Non-transacting supervisors/trainers excluded automatically (dont appear in CSV)

11. Excel Dashboard Push (Ecom v2.5)

Push to Dashboard button in Full Report header. Uses Microsoft Graph API OAuth2 implicit flow.

Target File: DC499-Dashboard.xlsx

- Location: DC499 Reporting > Excel folder on SharePoint/OneDrive
 - Drive ID: b!BqqqiF0owE6ponKYGHAIpZ0HJdD8w7ZKgyhbu8iJ0WISxRr3W4E1RIskB-MxqR9-
 - Item ID: 01DRZREETI3KCJ34LCMFHJ5TUMKY4QREIE
 - Status: Azure AD app registration pending IT approval
 - On first push: Creates EcomLog sheet and Dashboard sheet automatically
 - Each subsequent push: Appends one row to EcomLog, Dashboard updates via formulas
-

12. Code Patterns

Sorting — requires both TX ID AND Criteria column

```
var filtered = fullReportRows.filter(function(r) {
  if (txList.indexOf((r['Transaction ID']||'').trim()) === -1) return false;
  if (typ === 'sorting') return (r['Criteria']||'').trim() === SORT_CRITERIA;
  return true;
});
```

Roster format — always objects, never strings

```
// CORRECT:
```

```
return defaults.map(function(m) { return {email: m.email, enabled: m.enabled}; });
// WRONG – wraps object in object:
return defaults.map(function(email) { return {email: email}; });
```

onclick with dynamic keys — use dataset, not inline quotes

```
// CORRECT:
html += '<tr data-empkey="'+empKey+'" onclick="toggleAssocDetail(this.dataset.empkey)">';
// WRONG – causes SyntaxError:
html += '<tr onclick="toggleAssocDetail(''+empKey+'')">';
```

13. GitHub Deployment

Push files to main branch. GitHub Pages redeploys automatically (~60 seconds).

Git Workflow (with Claude Code)

```
git add .
git commit -m "descriptive message e.g. Ecom v2.6 – location filter, version bump"
git push
```

HTML files are 85-115KB. GitHub MCP connector times out on files ~95KB+. Use Claude Code git push or GitHub web UI drag-and-drop for large files.

Versioning — Update in 3 Places

- Settings footer: nordstrom-tag paragraph — vX.X
- Menu card badge: card-version span — vX.X
- Menu openApp call: openApp("Ecom_vX_X.html")

Change Type	Version Bump
Bug fix, small feature, logic change	Patch: v2.5 to v2.6
New major feature (Full Report expansion)	Minor: v1.9 to v2.0

14. Disclaimer Footer

All dept apps include this below the nordstrom-tag:

```
<p class="nordstrom-tag" style="margin-top:4px;max-width:560px;
margin-left:auto;margin-right:auto;line-height:1.5;">
  Disclaimer: This tool measures throughput only and may not be used to
  evaluate, coach, or hold team members accountable on performance.
</p>
```

15. Pending Work

- Ecom version bump to v2.6 — location filter was a meaningful data accuracy change
- Remove individual Replen/Putaway tabs from Ecom — Full Report is now standard
- PPH goal projection in Ecom Full Report (formula and values in Section 10)

- Roster drag-and-drop in Settings tabs (CSV/JSON, saves to localStorage)
- Date filter toggle — today only to prevent multi-day CSV inflation
- Shift card / Teams post — copyable shift summary card (UI designed, not built)
- Teams Incoming Webhook — post shift update to channel from Full Report button
- Azure AD app registration — IT ticket submitted for Graph API write access
- MegaDash integration — parked until live dashboard architecture resolved
- Power Automate folder-watch + auto-push — experimental, spec written
- changelog.json — update with all changes since v1.0 (after go-live)
- CLAUDE.md — project memory file for Claude Code sessions

16. Roadmap (from A3 Presentation)

Phase	Description	Status
Stabilization	Consistent roster and dashboard structure	Largely complete
Expansion	Automate workflow using Power Automate	In progress
Enhancement	Weekly trackers to compare workflow over several shifts	Pending
Scaling	Centralized hub across all departments	Pending